

HARSH KUMAR

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Education

Ajay Kumar Garg Engineering College

Bachelor of Science in Computer Science and Engineering (CSE) – CGPA: 7.5*

Oct. 2023 – Present

Ghaziabad, Uttar Pradesh

Jawahar Navodaya Vidyalaya, Hathras

Intermediate (Class XII) – Percentage: 82%*

April 2021 – March 2022

Hathras, Uttar Pradesh

Technical Skills

Languages: Python, C, HTML/CSS, JavaScript, SQL, Embedded C

Developer Tools: VS Code, Android Studio, Git, GitHub, Jupiter Notebook, Pycharm

Libraries: Numpy, Pandas, Matplotlib, SciPy, Scikit-learn

Frameworks: Django, TensorFlow, PyTorch

Robotics : Embedded Systems, Sensor integration, Microcontrollers, Actuator control, Computer vision, Debugging

Projects

SyncOrbit | Python, Django, Skyfield, scikit-learn, Leaflet, RTL-SDR, SQLite 🐙 GitHub

January 2026

- Developed an ML-driven predictive satellite handover optimization system to ensure seamless communication in LEO/MEO satellite networks.
- The system is designed to resolve one of the most significant operational challenges in LEO and MEO satellite architectures.
- Implemented real-time satellite tracking using Skyfield with live TLE data to compute orbital positions, subpoints, and visibility windows.
- Designed an ML-based RSSI degradation prediction model using scikit-learn to trigger proactive handovers before signal loss.
- Built a zero packet-loss buffering mechanism to maintain uninterrupted data transmission during satellite transitions.
- Created an interactive web dashboard using Django and Leaflet to visualize satellite positions, signal strength graphs, and handover events in real time.
- Integrated optional RTL-SDR hardware support for real RF signal sampling and spectrum analysis to simulate real-world satellite communication conditions.
- Structured the system using a modular architecture (orbit engine, signal model, ML engine, handover controller, buffer manager) to ensure scalability and maintainability.
- Implemented REST APIs for real-time satellite state updates, signal metrics, and handover control logic.

Automatic Waste Segregation System | Python, YOLOv8, OpenCV, ESP32, Siemens S7-1200 PLC

October 2026

- Designed and developed an AI-driven industrial waste segregation system integrating computer vision with electropneumatic actuation.
- Trained and deployed a custom YOLOv8 model for real-time classification of plastic and paper materials.
- Engineered a dual-conveyor system with synchronized motor control using ESP32 microcontroller.
- Integrated inductive metal sensors for early-stage metal segregation.
- Established communication between Python ML pipeline and Siemens S7-1200 PLC using Snap7 to actuate pneumatic cylinders.
- Implemented energy-efficient auto-shutdown logic based on conveyor inactivity detection.

Achievements

Achieved recognition in various national-level and collegiate competitions, showcasing skills in innovation, programming, and teamwork.

- Overall Winner(Software Track) of Hacksagon 2026 Hackathon at IIITM Gwalior.
- Winner of the HyperSpace Innovation Hackathon 2026 - A national level deep tech innovation Hackathon.
- Winner of the TechNova 2026 Hackathon at HRIT University.
- Winner of the GDG Hacker Cup 2026 Hackathon at KNIT Sultanpur.
- Winner of the Nationbuilders Ideathon 2026 at BCIIT Delhi.
- Participated in the ROBOCON 2025 national Finals.